

Notes

Types of errors: Syntax Error, Logical Error , Runtime Error

Syntax Errors (Compilation Error or Grammatical Error): Writing mistakes such as missing semicolon, missing brackets, extra brackets, using unnecessary symbols are examples of Syntax Error.

Only this error is detected at compile time by compiler

Logical Error: Writing wrong logic like, using + where – is needed, incorrect mathematical expression.

Runtime error (also known as Exception): These errors get detected at runtime. Dividing a number by 0, finding square root of a negative number – these are examples of run time error

Source Code: The High level code written by user is known as source code

Object Code: The Machine language Code (binary) we get after compiling the high level code is known as Object Code

Some languages use compilers , others use interpreter, Java uses both:

In java, the user written source code (High Level Code) is compiled by java compiler and byte code is created , this byte code is again given to java interpreter which converts the byte code into object code (Machine Level Code)

Source Code → Compiler → Byte Cde → Interpreter → Object Code

** Extension defines type of a file, for example music files are .mp3 , video files are .mp4 , MS WORD is .docx , powerpoint has .ppt , **Just like that extension for java source code is .java**

and extension for byte code is .class

Example: if a program has a class name “SUM” then the source code’s file name will be SUM.java and file name for byte code will be SUM.class

JVM: Java interpreter is known as JVM (JAVA Virtual Machine)

ASCII Table:

Character	ASCII CODE
A-Z	65-90
a-z	97-122
0-9	48-57
space (' ')	32